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Importance of screening for contrast sensitivity, falls, and mobility limitations in older adults with maculopathy

Catherine P Agathos [1,a], Natela M Shanidze [1], Donald C Fletcher [1]

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Abstract

Objective:

Assess the potential value of adding questions on falls/balance difficulties due to vision loss and testing contrast sensitivity in patients with maculopathy (mainly age-related macular degeneration, AMD) to help guide referral for mobility rehabilitation.

Design:

Cross-sectional study.

Participants:

125 patients over 55 years old with binocular maculopathy and intact peripheral vision presenting for a low vision rehabilitation consultation.

Methods:

Patients were asked questions regarding difficulties with their balance and mobility, including the 9-item Glaucoma Activity Limitation questionnaire (GAL-9). For analysis, patients were grouped by visual impairment severity using better-eye visual acuity (VA) and binocular contrast sensitivity (CS).

Main Outcome Measures:

Visual function, self-reported balance and mobility difficulties, falls history, and rehabilitation referrals were examined across visual impairment severity groups.

Results:

52% of patients reported balance or gait difficulties and 36% attributed mobility limitations to vision loss. Increasing vision deficit severity related to greater mobility concerns (>50% of patients in the severe categories rated difficulty as 3/5 or higher) and higher referral rates. Scores on the GAL-9 increased with greater vision loss and were predicted by contrast sensitivity, female sex, larger relative scotomata and falls history ($R^2 = 0.45$, $p < 0.0001$). CS was a better predictor of mobility difficulties than VA. Based on reported concerns, 31.2% of patients were referred to orientation and mobility training and/or physical therapy (>50% in the severe groups).

Conclusions:

Mobility limitations and falls affect a substantial proportion of AMD patients yet may go undiscussed in eye clinics. Incorporating screening questions and explicit discussion of mobility difficulties in clinical practice in maculopathy can help elucidate patient mobility limitations. Realization of these limitations is the prerequisite to initiate appropriate referrals to low vision and mobility specialists to improve mobility and manage falls risk. Clinical practice may also benefit from assessment of contrast sensitivity as this is a better indicator of difficulties in daily living. Our results, along with prior literature, underscore the need to recognize the role of central vision loss, and vision loss more broadly, in mobility decline. Thus, a

central vision loss-specific mobility questionnaire may be needed to facilitate patient screening in the future.

Keywords: central field loss, age-related macular degeneration, screening, falls, mobility, aging, contrast sensitivity

Introduction

Falls are a major and growing concern as they are the second leading cause of unintentional injury deaths worldwide[1] and the mortality rate of falls in individuals over 65 is increasing[2]. Falls among older adults are dangerous and costly: in the US alone, falls-related costs are projected to increase to $100 billion by 2030[3].

Visual impairment impacts mobility, increases fall risk, and is most prevalent among older adults[4–6]. Age-related visual impairments due to peripheral retinal damage, (e.g., glaucoma), are considered significant contributors to mobility limitations[7,8] given that peripheral vision is important for motion perception[9,10], spatial orientation[11], and balance and locomotion[12,13]. The association between falls and *central* visual impairment (e.g., age-related macular degeneration, AMD), however, is less emphasized clinically[14,15], despite a body of work revealing mobility difficulties[16–21], increased fall risk[22–24] and fall-related anxiety[25–28] in these patients, all of which reduce quality of life, and increase incidence of depression and mortality[29]. AMD is typically associated with and assessed in terms of loss of high acuity vision[30–33], as patients usually notice visual impairment when they experience difficulty with high-acuity tasks[34]. As a result, these tasks represent primary complaints[35] and falls are typically not considered a priority in AMD care[36]. Additionally, although central visual field loss is associated with decline in contrast sensitivity[34,37], which is important for successful performance of daily tasks including balance and mobility[19,38–46], standard clinical practice for maculopathy does not include contrast sensitivity assessment[47]. Given that AMD is the most common cause of irreversible age-related visual impairment in industrialized countries[48], affecting nearly 200 million individuals worldwide[49] and increasing with the aging population[50]; failure to systematically consider its effects on balance and mobility can lead to significant consequences for the autonomy, safety, and longevity of this population.

This study aimed to assess the potential value of a short falls/balance and mobility screening during low vision consultations for the management of central visual field loss. Additionally, given the lack of a central field loss-specific equivalent, we used the Glaucoma Activity Limitation questionnaire (GAL-9[51]) to further probe patients' mobility difficulties. Finally, in addition to assessing patients' self-reports of balance and mobility concerns relative to the severity of their vision loss, we also report resultant patient referral rates for mobility rehabilitation.

Methods

Participants

This cross-sectional study was reviewed by the Smith-Kettlewell Eye Research Institute's (SKERI) Institutional Review Board (IRB) and conforms with the the tenets of the declaration of Helsinki and the Health Insurance Portability and Accountability Act (HIPAA) regulations. Questionnaire, visual function, and demographic data were collected for 125 patients with maculopathy (26 males) who visited DCF's clinical practice at the Bicknell Envision Vision Rehabilitation Center in Wichita, KS over a one-year period. Inclusion criteria were the presence of binocular maculopathy and an intact peripheral visual field. Participants represent patients who were referred to DCF's clinic for low vision rehabilitation. Male and female sex was defined as having one or two X chromosomes respectively; in our sample, all participants had no more than two sex chromosomes. Prior to the start of their appointment, patients were asked for their consent to be asked additional (mobility-related) questions and for de-identified data collected during the appointment to be used for research and shared with the study team. All patients invited to participate agreed to do so. All patients with glaucoma, retinitis pigmentosa, neurological vision loss, diabetic retinopathy, or other peripheral vision loss were excluded. Patients presenting with other concomitant eye pathologies such as retinal vascular occlusions, cataracts, or other ocular media opacities were also excluded. Out of 125 patients, 9 had hereditary maculopathy (Stargardt's disease) and the other 116 presented with age-related macular degeneration. Maculopathy was confirmed via macular perimetry using the Scanning Laser Ophthalmoscope (MAIA, CenterVue, Fremont, CA, USA) monocularly in each eye. Intact peripheral visual fields were confirmed using confrontation visual field testing and medical history.

Visual function testing and visual impairment severity grouping

Visual acuity (VA) was measured monocularly in each eye, with habitual correction, using the ETDRS chart at 0.5, 1.0, or 2.0 meters (according to individuals' abilities). Contrast sensitivity (CS) was measured binocularly, with habitual correction, at 30 cm using the LEA symbols test[52], which provides figures between 25% and 0.6% contrast in six increments with score equivalents of 0.6 to 2.2 logCS. The LEA test is used in DCF's clinical practice as it does not require familiarity with the roman alphabet, thus accommodating a wide range of ages and linguistic backgrounds. The California Central Visual Field Test (CCVFT)[53] was used to estimate dense and relative binocular scotoma sizes. Horizontal and vertical scotoma diameters were averaged.

We examined questionnaire responses against vision metrics along a continuum and by grouping individuals according to visual impairment severity. Visual impairment severity as defined by the International Classification of Diseases (ICD) is based on visual acuity of the better eye and influences patient care and eligibility for benefits/legal rights. Given contrast sensitivity's higher relevance to mobility and daily living tasks[19,38–46], we also categorized visual impairment severity based on binocular CS. For VA-based grouping, we used standard WHO ICD-11 cutoffs[54] of mild: VA < 0.54 logMAR (< 20/70), moderate: VA = 0.54 – 1.0 logMAR (20/70 to 20/200), and severe acuity vision loss: VA >1.0 logMAR (>20/200). For contrast sensitivity-based grouping, we classified our participants based on prior work[55]; mild: CS ≥ 1.6 logCS (≤2.5% contrast), moderate: 1.30 logCS (5%), and severe contrast vision loss: CS ≤1.0 logCS (≥10%).

Mobility-related Questions and Referrals

Patients were asked general questions relating to their mobility and balance (Table 2, G1–5), followed by nine questions pertaining to activity-specific task scenarios (GAL-9[51]), to which participants reported their degree of difficulties on a 5-point Likert scale (Table 2, Q1–9). We chose the GAL-9 given its focus on mobility scenarios in daily life for visually impaired individuals (albeit with glaucoma), and its short length – key consideration for an in-consultation screening tool. While the GAL-9 was developed for peripheral vision loss, most mobility concerns it assesses are common across low vision. We used this tool to highlight mobility-related issues often overlooked in individuals with maculopathies in the clinic. Developing and validating a maculopathy-specific analog to the GAL-9 was beyond the scope of the current study.

Table 2.

Balance- and mobility-related questions used in the study

	Question	Response Format				
G1	How many times have you fallen in the past 12 months?	# of falls				
G2	Do you have problems with balance or gait? (e.g.., decreased balance or difficulty walking etc.)	Yes/No				
G3	Do you have mobility problems because of your vision?	Yes/No				
G4	Do you have mobility problems for other reasons? (i.e., do you have issues with moving around that are not related to vision, e.g., arthritis)	Yes/No				
G5	Have you received O&M training?	Yes/No				
How much difficulty does your vision give you, even with glasses, with the following activities?						
① None ② A little ③ Some ④ Quite a lot ⑤ Severe						
Q1	Walking after dark/dusk	①	②	③	④	⑤
Q2	Seeing at night	①	②	③	④	⑤
Q3	Walking on uneven ground	①	②	③	④	⑤
Q4	Adjusting to dim lights	①	②	③	④	⑤
Q5	Going from light to dark room and vice versa	①	②	③	④	⑤
Q6	Seeing objects come from the side	①	②	③	④	⑤
Q7	Walking on steps/stairs	①	②	③	④	⑤
Q8	Judging distance of foot to step/curb	①	②	③	④	⑤

	Question	Response Format				
Q9	Finding dropped objects	①	②	③	④	⑤

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As part of DCF's standard practice, all patients are referred to occupational therapy (OT). Referrals to orientation and mobility training (O&M) and/or physical therapy (PT) were made on a case-by-case basis for patients with significant mobility/balance complaints, and considering their insurance coverage and other factors influencing their ability to follow-up with the referral (e.g., transportation).

Analysis

This study's outcomes measures were patients' responses on the falls and mobility questionnaires (Table 2) and whether they obtained referrals. All outcomes were examined across the entire patient sample and grouped by visual impairment severity. The total score of responses on the GAL-9 was also examined in relation to the covariates of age, sex, visual function metrics, and number of falls. The GAL-9 total score was calculated as the sum of individual question scores (thus with a maximum score of 45). Self-reports on falls, balance, and mobility difficulties were examined in terms of frequency. Statistical relationships were examined using Spearman correlations. Normality was tested using the Shapiro-Wilk test. We used a generalized linear model of the GAL-9 total score with age, sex, visual acuity in the better and worse eye, binocular contrast sensitivity, binocular relative and dense scotoma size estimates, and number of falls (G1, Table 2) as the initial set of predictor variables to identify key factors in mobility limitations of our participant sample (Full Model in Table 3). The model was optimized using Akaike information criterion (AIC) estimation and retaining variables that did not present high collinearity (Preferred Model in Table 3). Alpha level was set at 0.05. All statistical analyses were performed using Prism 10 software (GraphPad Software, LLC, Boston, USA).

Table 3.

Linear regression model results for the full and simplified (preferred) models.

	Predictor	F (DFn, DFd)	Estimate (β)	95% CI	VIF	P-value
Full model	Intercept	F (8, 116) = 12.84 (regression)	29.80	15.96 to 43.64		**<0.0001**
	Age	F (8, 116) = 12.84	29.80	−0.259 to 0.005	1.189	0.0594
	Visual Acuity (better eye)	F (1, 116) = 3.63	−0.127	−2.625 to 4.575	1.716	0.5926
	Visual Acuity (worse eye)	F (1, 116) = 0.29	0.975	−2.588 to 6.035	1.918	0.4302
	Contrast Sensitivity	F (1, 116) = 0.63	1.723	−10.09 to −2.777	2.244	**0.0007**
	Binocular Dense Scotoma	F (1, 116) = 12.14	−6.434	−1.097 to −0.028	9.031	**0.0393**
	Binocular Relative Scotoma	F (1, 116) = 4.35	−0.563	0.196 to 1.012	8.647	**0.0041**
	Falls 12 months	F (1, 116) = 8.59	0.604	0.359 to 1.706	1.022	**0.0030**
	X chromosomes	F (1, 116) = 15.50	5.449	2.708 to 8.191	1.125	**0.0001**
Preferred Model	Intercept	F (5, 119) = 19.44 (regression)	31.20	18.58 to 43.82		**<0.0001**
	Age	F (1, 119) = 1.88	−0.088	−0.216 to 0.039	1.093	0.1734
	Contrast Sensitivity	F (1, 119) = 23.78	−7.580	−10.66 to −4.503	1.573	**<0.0001**

	Predictor	F (DFn, DFd)	Estimate (β)	95% CI	VIF	P-value
	Binocular Relative Scotoma	F (1, 119) = 6.47	0.232	0.0513 to 0.412	1.667	**0.0123**
	Falls 12 months	F (1, 119) = 10.08	1.077	0.405 to 1.749	1.005	**0.0019**
	X chromosomes	F (1, 119) = 15.62	5.382	2.685 to 8.078	1.076	**0.0001**

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Full model: DF = 116; R^2 = 0.47; Probability Correct = 23.7%.
Preferred Model: DF = 119; R^2 = 0.45; Probability Correct = 76.3%; AICc Difference = −2.34.
DFn/DFd: degrees of freedom numerator/denominator; CI: confidence interval; VIF: variance inflation factor

Results

Patient Characteristics

125 patients participated in this study, including 26 males. Participant ages ranged from 56 to 98 years (average: 80.6 ± 8.7). There were no sex differences in terms of age (Kolmogorov-Smirnov test, p=0.14) and number of falls in the past year (Fisher's exact test, p=0.43). Summary visual function data are presented in Table 1 along with demographics for the entire sample and according to severity of visual impairment. In our sample, 30.4% of patients had mild, 47.2% had moderate, and 22.4% had severe deficit, as defined by visual acuity. Grouping by contrast sensitivity, 31.2% of participants had mild deficit[b], 46.4% had moderate, and 22.4% had severe vision deficit. The proportion of patients grouped in the same level of visual impairment severity based on VA and CS was 59% in the mild, 60.3% in the moderate, and 42.9% in the severe category. There were no age differences between categories for either grouping (one-way ANOVA, p>0.2).

Table 1.

Summary demographics and visual function data for all patients and grouped by visual impairment severity using visual acuity (VA) and contrast sensitivity (CS).

		Age (years)	Years since diagnosis	VA in better eye (logMAR)	VA in worse eye (logMAR)	BinocularCS (log(CS))	Dense Scotoma Diameter (°)	Relativ Scoton Diamet (°)
	Total N=125 (26 M, 20.8%)	80.6 ± 8.7	1.8 ± 1.7	0.75 ± 0.34	1.21 ± 0.38	1.37 ± 0.43	5.0 ± 5.9	8.9 ± 7
VA Loss	**Mild** N=38 (6M, 15.8%)	80.3 ± 8.3	1.4 ± 0.8	0.39 ± 0.10	1.06 ± 0.51	1.62 ± 0.41	2.3 ± 3.6	5.5 ± 4
	Moderate N=59 (14M, 23.7%)	81.2 ± 9.4	1.5 ± 1.4	0.74 ± 0.13	1.20 ± 0.30	1.37 ± 0.34	3.8 ± 4.7	7.5 ± 6
	Severe N=28 (6M, 21.4%)	79.8 ± 7.7	3.1 ± 2.4	1.25 ± 0.16	1.43 ± 0.20	1.04 ± 0.40	11.4 ± 6.3	16.4 ± 8.6
CS Loss	**Mild** N=39 (7M, 17.9%)	79.7 ± 7.9	1.7 ± 1.6	0.49 ± 0.21	0.93 ± 0.39	1.87 ± 0.26	1.8 ± 2.4	4.3 ± 3
	Moderate N=58 (14M, 24.1%)	81.5 ± 8.7	1.8 ± 1.3	0.83 ± 0.30	1.29 ± 0.30	1.30 ± 0.00	5.4 ± 6.3	8.9 ± 7

		Age (years)	Years since diagnosis	VA in better eye (logMAR)	VA in worse eye (logMAR)	BinocularCS (log(CS))	Dense Scotoma Diameter (°)	Relativ Scoton Diamet (°)
	Severe N=28 (5M, 17.9%)	80.2 ± 9.8	2.1 ± 2.3	0.93 ± 0.36	1.44 ± 0.30	0.82 ± 0.27	8.8 ± 6.3	15.4 ± 7.8

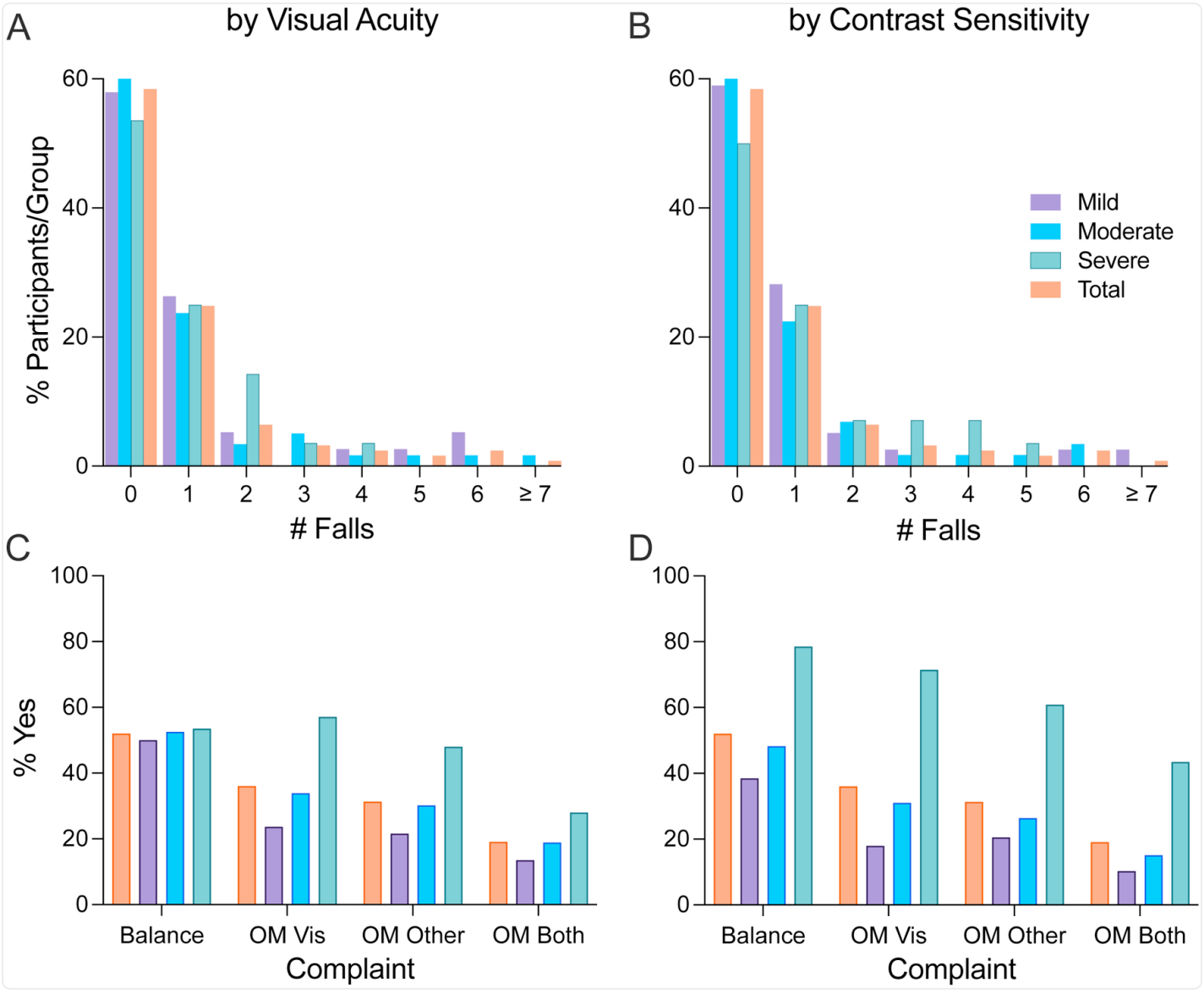
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Mean values are provided with standard deviation. VA: visual acuity; CS: contrast sensitivity; M: male participants. Dense and relative scotoma size are binocular estimates based on the CCVFT[53].

Self-Reports of Falls, Balance, and Mobility Complaints

41.6% of patients reported having at least one fall in the previous twelve months. The maximum reported number of falls was 10 (one participant), and the average was 1 fall/person within each severity group (regardless of classification type – VA or CS). Figure 1 A&B shows a detailed breakdown of the number of falls across the cohort (peach bars) and separated into the three severity groups. While there was a higher ratio of patients with severe impairment (teal) who had 2–4 falls, several participants in the mild (violet) and moderate (blue) groups had the highest number of falls in a year.

Figure 1. Patient reports of number of falls in the past twelve months and balance and mobility problems (C, D) for the total cohort (peach) and by disease severity based on visual acuity (A, C) and contrast sensitivity (B, D).



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Fall frequency (**A, B**) was similar among groups, with the most recurrent fallers emerging in the mild (violet) and moderate (blue) groups. **C, D:** Percentage of patients reporting difficulties with balance and gait (*Balance & Gait*), and mobility due to their vision (*Mobility Vision*), due to other factors (*Mobility Other*), and their overlap (*Mobility Both*). Note that grouping by contrast

sensitivity (D) reveals a greater proportion of patients in the severe category (teal) reporting difficulties, compared to those in the mild (violet) and moderate (blue) categories.

In our sample, 52% of patients reported having balance or gait problems ("Balance & Gait" in Figure 1, C&D). This incidence was closely maintained across severity groups when grouped by visual acuity of the better eye: 50.0% of participants in the mild, 52.5% in the moderate, and 53.6% in the severe impairment categories complained of balance or gait issues. When grouped by binocular contrast sensitivity, however, incidence of reported balance problems increased with severity: 38.5% in the mild; 48.3% in the moderate, and 78.6% in the severe participant group.

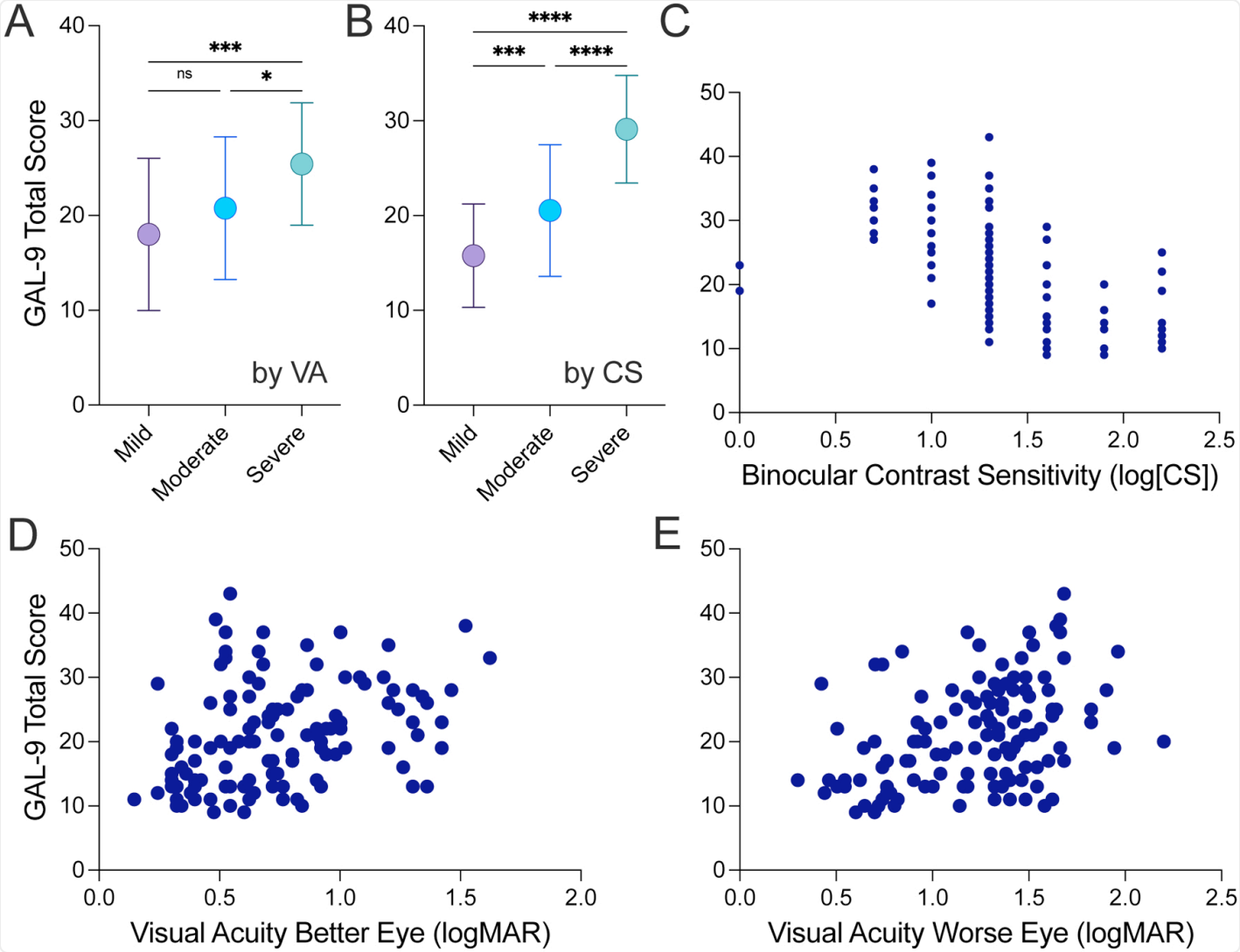
36% of patients reported mobility difficulties which they attributed to vision deficits ("Mobility Vision" in Figure 1C&D), compared to 31.3% with non-vision-related mobility difficulties ("Mobility Other" in Figure 1C&D), with a 19.1% overlap ("Mobility Both" in Figure 1C&D). When grouped by VA, 23.7% in the mild, 33.9% in the moderate, and 57.1% in the severe groups complained of mobility difficulties due to their vision deficit. When grouped by CS, 18.0% in the mild, 31.0% in the moderate, and 71.4% in the severe groups complained of vision deficit-related mobility difficulties.

When grouped by VA, 21.6% in the mild, 30.2% in the moderate, and 48.0% in the severe groups said they had mobility difficulties that were not related to vision loss. When participants were grouped by CS, 20.5% in the mild, 26.4% in the moderate, and 60.9% in the severe groups reported mobility difficulties that were not vision related.

Activity limitation questionnaire responses

The average GAL-9 total score for the entire sample was 21.0 ± 7.9. Figure 2 A&B shows mean total scores on the GAL-9 questionnaire for patients grouped according to visual impairment severity. Mean total scores were significantly higher in patients with severe as opposed to moderate and mild visual impairment, whether severity grouping was based on VA or CS (one-way ANOVA with Tukey adjustment for multiple comparisons for **VA-grouping**: $F(2,97) = 8.05$, $p < 0.001$; mean scores per VA group: $\text{mean}_{\text{mild}} = 18.0 \pm 8.0$; $\text{mean}_{\text{moderate}} = 20.8 \pm 7.5$; $\text{mean}_{\text{severe}} = 25.4 \pm 6.5$; for **CS-grouping**: $F(2,97) = 37.47$, $p < 0.0001$; mean scores per CS group: $\text{mean}_{\text{mild}} = 15.8 \pm 5.5$; $\text{mean}_{\text{moderate}} = 20.5 \pm 6.9$; $\text{mean}_{\text{severe}} = 29.1 \pm 5.7$). Additionally, when grouped by CS, the scores of patients with moderate vision loss were also significantly higher than those of patients with mild vison loss.

Figure 2. Mean total score on the GAL-9 questionnaire for each visual impairment severity group and relationship to visual function.



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Mean total scores based on visual acuity **(A)** and contrast sensitivity **(B).** In both cases, mean total scores were significantly higher in patients with severe as opposed to moderate and mild visual impairment. When grouped by contrast sensitivity, the difference in total scores between moderate and mild groups was also significant. Relationship between total score on the GAL-9 and binocular contrast sensitivity **(C),** visual acuity of the better eye **(D)** and worse eye **(E)** for each patient. Significant relationships were found for all three metrics indicating a higher score associated with worse visual function ($p < 0.0001$). **A**: *: $p = 0.02$; **: $p = 0.0003$; ns: not

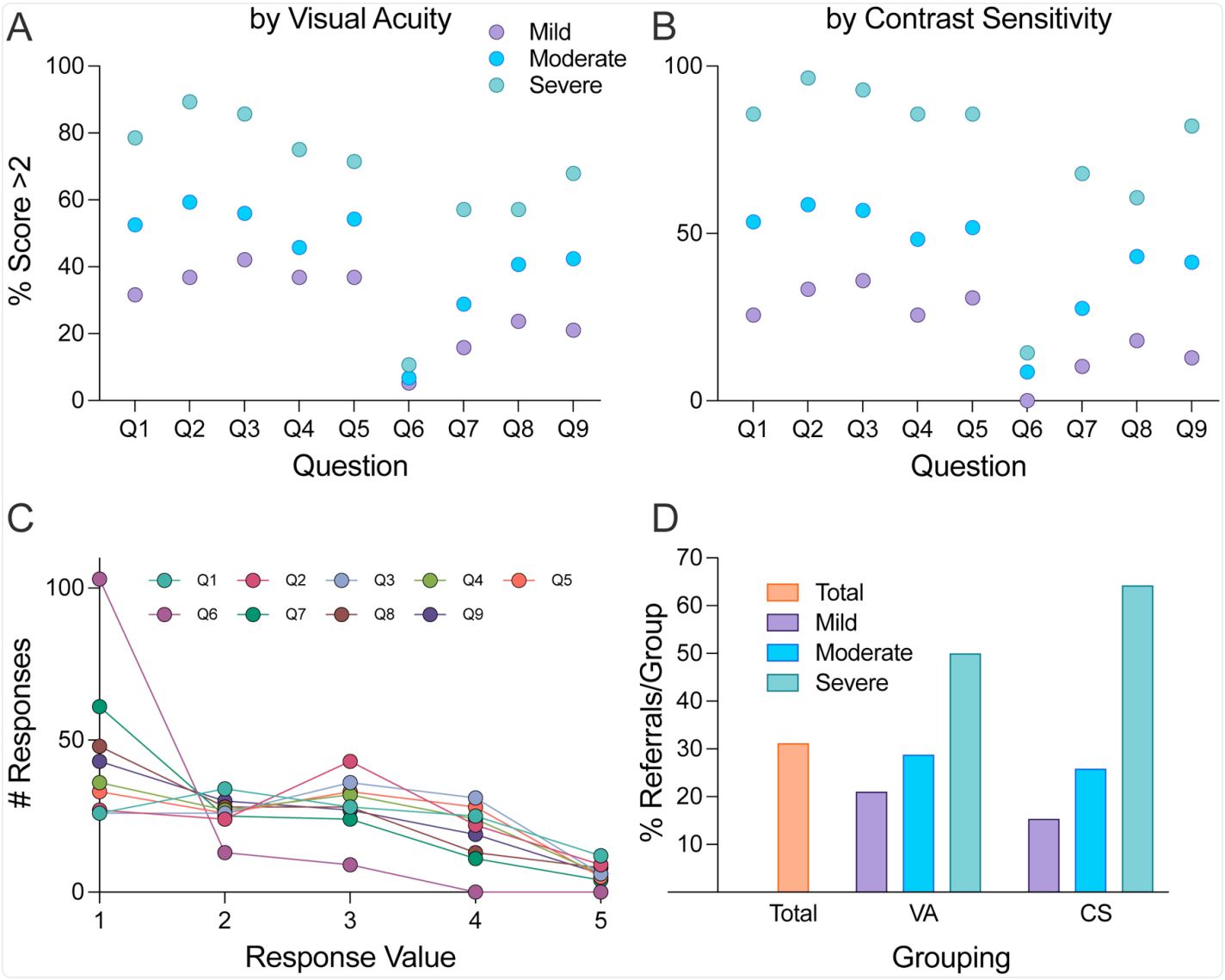
significant. VA: visual acuity; CS: contrast sensitivity. **B**: ***: $p = 0.001$; ****: $p < 0.001$. Error bars: standard deviation.

Next, we examined the GAL-9 total scores against patients' visual function metrics and number of falls in the previous year. Figure 2 also shows the relationship between the total score on the GAL-9 and visual acuity (Figure 2D&E) and contrast sensitivity (Figure 2C) metrics for all patients. We found significant relationships for all three metrics with a higher score being associated with worse visual function (**better eye VA**: Spearman $\rho = 0.39$, $p < 0.0001$, 95% CI: 0.22 — 0.53; **worse eye VA**: $\rho = 0.37$, $p < 0.0001$, 95% CI: 0.21 — 0.52; **binocular CS**: $\rho = -0.62$, $p < 0.0001$, 95 % CI: −0.72 — −0.49).

To determine the best set of predictors of the GAL-9 total response score, we used linear regression modeling with age, sex, better and worse eye visual acuities, binocular contrast sensitivity, relative and dense binocular scotoma estimates, and the number of falls in the last 12 months as the initial set of predictors. We simplified the model using AIC prediction error estimation ($R^2 = 0.45$, $p < 0.0001$). While both relative and dense binocular scotoma estimates were identified as significant predictors, they were highly correlated leading to elevated variance inflation factors (see VIFs in Full Model of Table 3), only the relative scotoma estimate was included in the simplified model as this was associated with the best AIC score. The significant predictors in the optimized model were: sex ($\beta = 5.38$, $p = 0.0001$), contrast sensitivity ($\beta = -7.58$, $p < 0.0001$), relative binocular scotoma size ($\beta = 0.23$, $p = 0.0123$) and falls ($\beta = 1.08$, $p = 0.0019$). Age was also included in the model but was not in itself significant ($\beta = -0.09$, $p = 0.173$). *Lower* contrast sensitivity, female sex, larger relative scotomata, and number of falls were linked to greater mobility-related activity difficulties (higher GAL-9 total). The model results (full and AIC preferred) are presented in Table 3.

Figure 3C shows the frequency distribution of patients' ratings of their difficulties for individual items on the GAL-9. Activities relating to dark environments (Q1,2), lighting transitions (Q4,5), and uneven surfaces (Q3) were rated as most difficult. Q3 "walking on uneven ground" and Q6 "seeing objects from the side" were rated as the most and least difficult, respectively. The degree of difficulty reported increased with visual impairment severity, with more than 70% of patients in the 'severe' group rating most questions as 'somewhat' or more difficult (≥3/5, Figure 3A&B).

Figure 3. Patients' difficulty ratings on the GAL-9.



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The proportion of patients in each visual impairment severity group based on visual acuity (**A**) and on contrast sensitivity (**B**) scoring individual questions with a 3 out of 5 or higher (i.e., "some", "quite a lot" or "severe" difficulty) is shown for mild (violet), moderate (blue), and severe (teal) groups. **C.** The number of patients scoring each question (color-coded in the figure legend) on the 5-point Likert scale. **D.** The percentage of patients receiving referrals for O&M and/or PT in total and grouped by visual impairment severity (based on visual acuity and contrast sensitivity). More than 50% of patients in the severe groups were referred. Note the greater proportional differences between moderate and severe groups when using contrast

sensitivity as a severity classifier, as opposed to visual acuity in A, B and D. VA: visual acuity; CS: contrast sensitivity.

Referrals

A total of 31.2% of our cohort obtained referrals to mobility rehabilitation (Figure 3D). When grouped by acuity, 21.1% of patients in the mild, 28.8% in the moderate, and 50.0% in the severe groups were given referrals to O&M and/or physical therapy rehabilitation professionals. This breakdown when using contrast sensitivity for severity grouping was: 15.4% in the mild, 25.9% in the moderate, and 64.3% in the severe groups. Patients receiving referrals had significantly higher total GAL-9 scores ($U=373.5$, $p<0.00001$; $mean_{referred} = 28.5 \pm 6.9$, $mean_{not\text{-}referred} = 17.6 \pm 5.6$). Prior to the study, four patients in our sample had been referred to O&M (G5, Table 2; and thus did not obtain new referrals). All patients received referrals to occupational therapy as part of clinic's standard practice.

Discussion

Balance, mobility, and falls are clinically important issues in central field loss

The outcomes of this study highlight the importance of low vision care professionals addressing fall risk and mobility in individuals with central visual field loss (CFL) in the clinic, both to guide referrals and help patients reflect on the nature of these limitations. Participants' responses challenge the traditional view that low vision patients with an intact peripheral field are less likely to have balance and mobility limitations[15,56,57]. Over half of our sample reported balance problems, and many expressed mobility difficulties attributed to vision loss, adding to evidence suggesting that mobility limitation and fall-risk are major issues in CFL needing consideration in low vision care[15,22,23,58].

Screening patients by asking about fall history and balance/mobility complaints is an important first step in enhancing standard clinical practice, since patients, especially older adults, seldom spontaneously volunteer information that feeds into the stigma of disability[59] (and likely not to their eye-care provider). A mobility-focused questionnaire, complemented by visual function screening, can also inform appropriate referrals.

In line with prior work[28,23,24,60], 41% of our sample reported at least one fall in the previous year. The fact that only participants in the mild and moderate groups reported having 5 or more falls may reflect reduced activity in the severe group[16,25,26]. This relationship warrants further investigation as fewer falls

may also be masking the serious issues of isolation. Indeed, this confound is why a finer screening tool that addresses daily living tasks is needed.

Our findings suggest that AMD patients have similar mobility difficulties as those with glaucoma, as evidenced by the reported concerns on the mobility scenarios of the GAL-9 questionnaire (except, unsurprisingly, Q6). Total scores were also close to those reported by glaucoma patients in the literature[61]. While we do not attempt to validate the GAL-9 in AMD or suggest that it be used for screening in patients with maculopathies, our findings are in line with studies reporting comparable mobility issues, and fall risk and incidence in individuals with central and peripheral field loss[21,60,62]. The prevalence of falls (Figure 1A&B) and referral rates in the mild group (Figure 3D) suggest that the bar for balance/mobility rehabilitation needs to be lowered for all CFL patients – particularly since patients with better VA or monocular field loss are unlikely to be referred for low vision rehabilitation despite clear benefits[63,64].

Given the high prevalence of visual impairment and comorbidities/fall risk factors in older adults[6,59], inquiring about such issues by *all* clinicians is important for rehabilitation needs and recommended in the fall-risk literature[65]. Early identification of balance/mobility difficulties in the vision clinic is important as i) vision loss often progresses over time, likely compounding age-related mobility decline and ii) these difficulties tend to be attributed to aging[66,67] rather than visual impairment by both the patient and their provider. Indeed, in our cohort, most patients did not volunteer their concerns *prior* to being asked about balance and mobility.

Finally, research on fall risk and balance/motor control[68–72] suggests that balance/mobility deficits[18,19,29,73,74], higher incidence of falls[22–24,75], and concerns about falling[26,28,76] in CFL may also be due to a combination of visual and oculomotor deficits[77–79], visual dependence[80], and cognitive decline[74] in this population. Mobility rehabilitation professionals can leverage their expertise to address these problems in CFL patients following appropriate referrals from eye-care professionals.

Role of visual function in mobility concerns

In agreement with prior work[19,40,81], a key finding in our study is that contrast sensitivity is associated more strongly and is more discriminative of balance and mobility difficulties than visual acuity. These findings further underscore the clinical value of CS assessment for understanding functional vision in AMD (and more generally low vision), particularly in the context of fall risk[19,40,41]. Indeed, a number of studies highlight the better discriminative ability of CS measures, as opposed to VA[31,46,82], and its relation to performing activities of daily living, including balance/mobility[19,38–46]. Further, CS is a more sensitive marker of AMD progression and treatment effects than VA[37,83], making contrast sensitivity

testing especially valuable in clinical practice. In the present study, as in studies of peripheral field loss[51,84], we found that the GAL-9 total score was associated with both CS and VA. This commonality highlights that mobility is of broad concern across low vision.

Study limitations

There are several limitations to our study. First, while we do report on patients' balance and mobility concerns, we do not use an AMD-specific, validated questionnaire to do so. Instead, we rely on an instrument developed for assessment of patients with peripheral field loss, general mobility questions, and questions relating to falls. We chose this approach for several reasons. First, unlike for peripheral vision loss[51,84,85], there is currently no CFL-specific *mobility* questionnaire. The more general quality of life questionnaires validated for AMD, like the Impact of Visual Impairment[86] and the Macular Disease on Quality of Life[87] or general low vision questionnaires, such as the NEI-VFQ are longer and have a limited focus on mobility concerns, making their implementation in the clinic less practical. On the other hand, general balance and mobility-specific questionnaires like the Activities-specific Balance Confidence (ABC) scale[88] are not focused on vision-specific limitations. Thus, we chose the GAL-9, which was shown previously to be effective at detecting mobility concerns in glaucoma, with a good discriminative ability of disease stage[89]. Our findings suggest that those with AMD have similar mobility concerns to those with glaucoma[61] on the same set of tasks.

A further limitation is the disproportionate number of female to male patients in our sample. While there were no sex differences in terms of age and number of falls (see Methods), the regression models revealed that women were more likely to report mobility and balance concerns. Given the older age of our participants, the larger proportion of women in our sample is likely due to differences in lifespan (as illustrated by the demographics of our sample's region[90]). This trend is consistent with similar studies in macular degeneration[28], as well as what is observed in our clinic, where women outnumber men by a factor of 5 regardless of pathology.

Finally, it is important to note that all participants surveyed in this study were patients in a low vision rehabilitation clinic, having been referred there due to functional limitations. It is possible that if surveyed from general eye clinics, the patient population profile may be different both in terms of patient awareness of the functional deficits, as well as the deficits themselves. These potential differences bear investigation in future research.

Finally, recall bias may be a concern when self-reporting falls history, particularly as studies indicate that older adults tend to under-report their number of falls[91–93]. This is mostly due to misperceptions of what constitutes a fall or forgetting falls that were not injurious. This issue with self-reporting falls history is

also why it is encouraged to complement history with questionnaires regarding specific mobility or stepping situations, questionnaires on concerns about falling and objective balance and mobility assessments[92].

Conclusions and future directions for falls prevention in maculopathy patients

Contrast sensitivity and balance/mobility screening are key to enhancing CFL (and more broadly low vision) care. While the GAL-9 revealed mobility difficulties of maculopathy patients in our study, our research suggests the need for a mobility questionnaire validated specifically for CFL. Building on this study's and prior literature's findings that contrast sensitivity is most closely related to mobility limitations, other ways of mitigating fall risk could be through contrast enhancement. For example, refraction could be optimized using contrast enhancing filters and augmented reality devices, while indoor lighting can enhance scene contrast in different environments. Screenings like the one used in this study can also be refined and complemented with objective balance/mobility assessments like measuring walking speed[59] which is known to be affected by CFL[18,94], and is easy to implement in the clinic. Today, fall-risk identification and mitigation are feasible and promising. The development of enhanced screening and referral practices is a key next step in the care of individuals with central field loss.

Highlights.

- Balance and mobility difficulties are prevalent in individuals with maculopathies
- Screening via self-report in the clinic can reveal balance/mobility difficulties
- Inquiring about fall risk-related concerns can lead to appropriate referrals
- Contrast sensitivity correlates better than visual acuity with mobility issues

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Footnotes

[b]Note that contrast sensitivity thresholds above 1.6 logCS are not technically considered visual impairment[55]. For simplicity and consistency across comparisons with visual impairment severity according to visual acuity, we use the term “mild deficit” nonetheless.

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